

Full-Stack Developer Technical Exam

Laravel (API) + React + Python

Design and implement a system that extracts, stores, and displays posts from multiple local HTML files attached with this exam. The system must use a Python script to parse the HTML files and extract the post title, author, content, related images, and source file name. The extracted data should be stored in a MySQL database, exposed through a Laravel API, and presented through a React frontend.

The Python parser must not depend on any paid services, including paid scraping tools or third-party automation platforms. It should rely only on free and controlled methods for processing local HTML files.

The system should store the original post content without rewriting or changing it. It should prevent duplicate imports, safely handle malformed HTML files, and keep the scraping, database, API, and frontend layers cleanly separated. The Laravel API should support listing posts with pagination, retrieving a single post, searching, and filtering by category. The React frontend should display posts by category, provide search and filtering, and include a single post page with related images.

The final implementation should include the Python HTML parser, MySQL schema or Laravel migrations, Laravel API endpoints, React frontend components, and clear setup instructions for running the full system.

Submission Instructions

Submit a Git repository containing:

- README for the details and setup instructions
- Python script (data ingestion & classification)
- Laravel backend (API)
- React frontend
- Database migrations or SQL dump